Project 1 — The Arcade

HTML, CSS, and Design

Overview

This first assignment has you building the site that every other project this semester will be added to. By the end of the course it will hold four games; right now it holds one puzzle, plus enough about you that the page is something you can put on your resume as either a homepage or a demo of your skills.

You will build a small static site containing a landing page, a playable mini crossword, a page about your experience and education, and a page describing how to contact you. The crossword is the centerpiece and the reason this project is more than a personal site — you are going to build a real interactive grid using nothing but HTML and CSS.

Keep in mind who this is for. You will be sharing this URL with recruiters and colleagues, and you will be adding to it for the next three months. Build something you would be happy for someone to find under your name.

There is an emphasis on creativity and individuality here. If you have an idea that does not quite conform to the expectations below, contact the course staff for approval. You can also always change the site after the semester is over.

This project is to be done solo. Later projects will allow you to work in pairs.

Reminder: As long as you try your best, you will receive a B+ in this course.

Reminder: You should use AI to help explain, provide next steps or hints or receive feedback. You should not be using AI to generate this project or the writeup for you.

Restrictions

No JavaScript or code at all: no imported JS libraries, no Ruby, no other programming language, no logic of any kind on the page. Everything interactive in this project has to be done with HTML and CSS alone, which is the whole exercise.

No styling libraries either: no Bootstrap, no Tailwind, no Material, no component frameworks. You will lose points for these. The goal is for you to become comfortable writing CSS yourself before you let a tool write it for you.

You MAY import font files and icon files. You may use SVG. That is the extent of what may come from outside your own repository.

A Note on Hosting

GitHub Pages is recommended and is covered in lecture. You may use another host (Netlify, Cloudflare Pages, your own server) if you are comfortable with it, but the site must be genuinely static (ie, no code, server/client logic, etc).

Whatever you use, the URL must work for a stranger with no login.

Rubric

Item	Percent
Working GitHub repo, code quality, and a live URL	10
Four unique pages and file organization	10
Navbar	15
The puzzle	15
Mobile friendly design	10
Use of certain HTML elements	10
Use of certain CSS properties and features	10
Accessibility	10
Good and professional design	10
Writeup	Required
Extra credit	5 pts

Working GitHub Repo, Code Quality, and a Live URL

All of your code must be pushed, and the site must actually work for a visitor. Code should be clean, readable, and consistently formatted.

- Organize files into sensible folders. Shared CSS in a common location, page-specific CSS alongside its page
- The core HTML must be an index.html, and I should never see a .html in your URL paths
- Meaningful, consistent names for ids and classes. Carelessly named selectors will cost you points
- A .gitignore, committed before your first real commit
- More than one commit, with messages that describe the change
- No JavaScript files anywhere in the repository

Four Unique Pages

Your site needs at least four pages:

- Landing page: the arcade itself. It introduces the site and links to the puzzle. This is your index.html
- The puzzle page: covered in detail below
- About Me: your experience, education, and anything else professional you want visible
- Contact: how to reach you and where to find you online. Include what applies: email, GitHub, LinkedIn, and any social links whose content you are comfortable with professionals seeing

If you would rather have a fifth page (a blog with at least one entry, photography, hobbies, reading list), go ahead.

Content is not graded on typos or grammar, and if you would rather not publish real personal details you may make them up.

Navbar

Every page needs the same navigation.

- It stays in position no matter where the user has scrolled
- It links to every page on the site
- The current page is visually indicated, so a visitor always knows where they are
- It includes a title referencing your name
- It does not cover content unless the user has scrolled
- It is marked up as a nav containing a list of links — not a row of divs
- It is fully usable with the keyboard alone, with a visible focus indicator

Hint: position: sticky does most of the work. Consider using transitions and pseudo-classes to change the navbar as the user interacts with it.

The Puzzle

The centerpiece. You are building a playable mini crossword with no JavaScript whatsoever.

Requirements:

- A grid of at least 5×5 (though you can make it as big as you wish)
- Built with CSS Grid
- Blocked-out squares, visually distinct from open ones
- Clue numbers in the corner of the appropriate squares
- Across and Down clue lists, marked up as real lists
- Typeable cells — a visitor can click or tab into a square and enter a letter
- A way to reveal the solution that works for keyboard and touch users

Tip for input:

Set input with maxlength="1" in each open square to make it easy: it accepts a single letter, automatically tabs to the next square, and no need for JS.

```
<input type="text" maxlength="1" aria-label="Row 1, column 2, 3 Across">
```

Every input needs an accessible name. A grid of twenty-five unlabelled text boxes is unusable with a screen reader.

On revealing the solution:

Do not gate the solution behind :hover alone. Hover does not exist on a touchscreen and cannot be reached from the keyboard, so a hover-only reveal is invisible to phone users and keyboard users both. Use one of these instead:

- A details / summary element, which is semantic and keyboard-accessible with no tricks
- The checkbox technique — a checkbox plus a sibling combinator on :checked

```
input:checked ~ .solution { opacity: 1; }
```

You are welcome to add hover effects on top of that as decoration. Just do not make them the only way in.

You may write your own mini puzzle or adapt a public-domain one. If you adapt one, credit it in your writeup.

Mobile Friendly Design

The site must work as well on a phone as on a monitor. Assume the person deciding whether to interview you is looking at it on a phone.

- Everything reachable on desktop must be reachable on mobile
- Text, images, and the puzzle grid scale sensibly
- The navbar likely needs to change shape on a small screen — moving it to the bottom is one good option
- Tap targets are at least 44×44 pixels, including the puzzle squares
- Test from 320px wide up to a wide monitor. DevTools device mode is the fast way to do this

Use of Certain HTML Elements

Your site must use each of the following at least once:

- header, nav, main, footer
- section or article
- h1 through h6, used as a real outline — never chosen because you liked the size
- a, img (with meaningful alt text), p
- ul or ol
- label and input

Your HTML must pass the W3C validator with no errors. Semantic elements are expected wherever one applies; div and span are for when nothing else fits, not as a default.

Use of Certain CSS Properties and Features

Your stylesheets must use each of the following at least once:

- font-family
- background or background-color
- margin and padding
- position (your sticky navbar counts)
- display: grid — the puzzle
- display: flex
- @media queries
- Custom properties (CSS variables) for your colors and spacing, defined once at the top
- At least two different pseudo-classes
- At least one pseudo-element
- :focus-visible, styled deliberately

- transition or transform

Accessibility

The entire site is usable with a keyboard alone, and focus is always visible

- All text meets a contrast ratio of at least 4.5:1
- Every image has appropriate alt text, and decorative images use an empty alt
- Every input has an accessible name
- No clickable divs: use buttons or links
- Lighthouse accessibility score of 95 or higher

Run Lighthouse before you submit in your browser and will point out errors

Good and Professional Design

The site should be easy to navigate and interesting enough to hold attention.

- Decide on a type scale, a neutral color ramp, one accent color, and a spacing scale before you write CSS. Define them as custom properties
- Cap your body text at a readable line length
- Consider importing a font. Which typefaces do you actually find readable on a screen?
- Think about branding: SVG icons and a consistent visual identity go a long way
- Use spacing to group related things. Whitespace is doing work
- I don't want to see default colors, sizes, fonts, etc. There should be colors and an interesting visual component.

This is not a design course. The intention is that you can produce something worth sharing with recruiters and colleagues and start thinking about how design affects the UX.

Extra Credit

Submit 48 hours before the deadline and you earn 5 extra credit points.

Deliverables

Submit the following:

- A link to your live website
- A link to your GitHub repository. If it is private, add the TAs as collaborators
- A writeup answering the questions below, at least three sentences per bullet, where appropriate

Writeup questions:

- What was the most challenging part of this assignment? Did you find HTML and CSS easy or difficult to work with?
- How did you build the puzzle grid, and what did you try that did not work?
- Is a crossword grid tabular data? Make the case for marking it up as a table, then make the case for a grid of divs, and tell me which you chose and why
- What decisions did you make to get the site working on mobile?

- What did you take into account when designing the site? Is there anything you are particularly proud of?
- Your navigation will need to hold four games by the end of the semester. What did you do, if anything, to make that easier for yourself later?
- Given more time or resources, what would you add?
- How many hours did you spend on this assignment?
- (Optional) If you made any assumptions about this assignment, what were they?
- If you used code or design from anywhere online, say so here. If you imported a font or icon library, or adapted an existing puzzle, note that as well